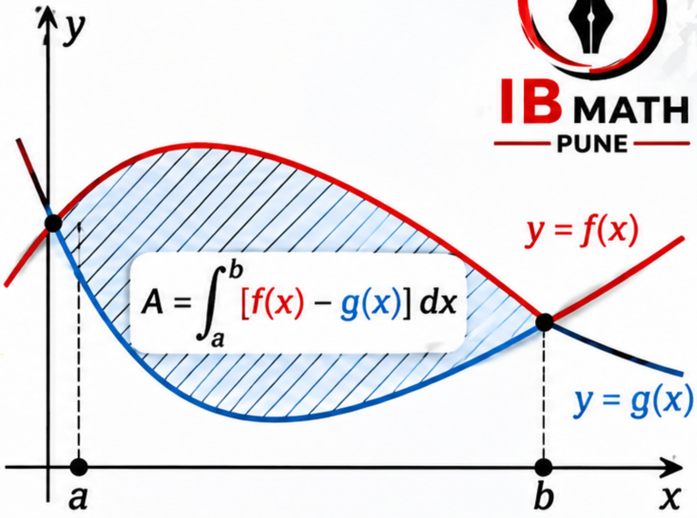
HOW TO FIND THE AREA BETWEEN TWO CURVES IN 5 EASY STEPS

FOR IB AAHL & AIHL

IB MATH PUNE



1. FIND POINTS OF INTERSECTION
2. DETERMINE THE LIMITS
3. IDENTIFY UPPER & LOWER CURVES
4. SET UP THE INTEGRAL
5. EVALUATE THE INTEGRAL

PERFECT FOR IB EXAMS!

AAHL

SL 5.5

Content	Guidance, clarification and syllabus links
Introduction to integration as anti-differentiation of functions of the form $f(x) = ax^n + bx^{n-1} + ...$, where $n \in \mathbb{Z}$, $n \neq -1$	Students should be aware of the link between anti-derivatives, definite integrals and area.
Anti-differentiation with a boundary condition to determine the constant term.	**Example:** If $\frac{dy}{dx} = 3x^2 + x$ and $y = 10$ when $x = 1$, then $y = x^3 + \frac{1}{2}x^2 + 8.5$.
Definite integrals using technology. Area of a region enclosed by a curve $y = f(x)$ and the x -axis, where $f(x) > 0$.	Students are expected to first write a correct expression before calculating the area, for example $\int_2^6 (3x^2 + 4)dx$. The use of dynamic geometry or graphing software is encouraged in the development of this concept.

SL 5.11

Content	Guidance, clarification and syllabus links
Definite integrals, including analytical approach.	$\int_a^b g'(x)dx = g(b) - g(a)$. The value of some definite integrals can only be found using technology. **Link to**: definite integrals using technology (SL5.5).

Content	Guidance, clarification and syllabus links
Areas of a region enclosed by a curve $y = f(x)$ and the x-axis, where $f(x)$ can be positive or negative, without the use of technology. Areas between curves.	Students are expected to first write a correct expression before calculating the area. Technology may be used to enhance understanding of the relationship between integrals and areas.

AHL 5.17

Content	Guidance, clarification and syllabus links
Area of the region enclosed by a curve and the y-axis in a given interval. Volumes of revolution about the x-axis or y-axis.	

AIHL

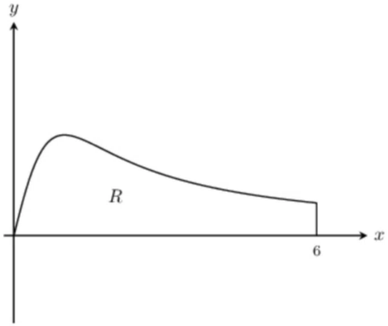
AHL 5.12

Content	Guidance, clarification and syllabus links
Area of the region enclosed by a curve and the x or y-axes in a given interval.	Including negative integrals.
Volumes of revolution about the x- axis or y- axis.	$V = \int_a^b \pi y^2 dx$ or $V = \int_a^b \pi x^2 dy$

Question (1):

[Maximum mark: 6]

The following diagram shows the graph of $f(x)=\dfrac{4x}{x^2+1}$, for $0 \le x \le 6$, and the line $x=6$.



Let R be the region enclosed by the graph of f, the x-axis and the line $x=6$.

Find the area of R.

$$R=\int_0^6 \frac{4x}{x^2+1}dx = 2\int_0^6 \frac{2x}{x^2+1}dx$$

Let $u=x^2+1 \quad \therefore du = 2xdx$

$$\therefore R = 2\int \frac{du}{u} = 2\ln|u| = 2\ln\left|x^2+1\right| + C$$

$$\therefore R = 2\left[\ln\left|x^2+1\right|\right]_0^6$$

$$= 2\{(\ln 37) - \ln(1)\}$$

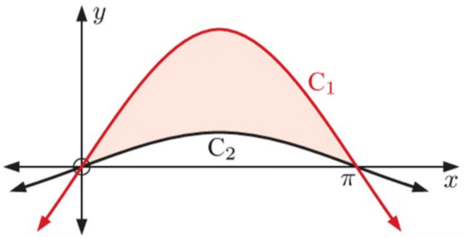
$$= 2\ln 37 - 0 \longrightarrow R = 2\ln 37 \text{ units}^2$$

Question (2):

The illustrated curves are the graphs of $y = \sin x$ and $y = 4\sin x$.

a Identify each curve.

b Calculate the shaded area.



(a) $y = \sin x$ has amplitude 1

and $y = 4\sin x$ has amplitude 4

$\therefore c_1$ is $y = 4\sin x$

and c_2 is $y = \sin x$

Therefore area A between 0 and π is

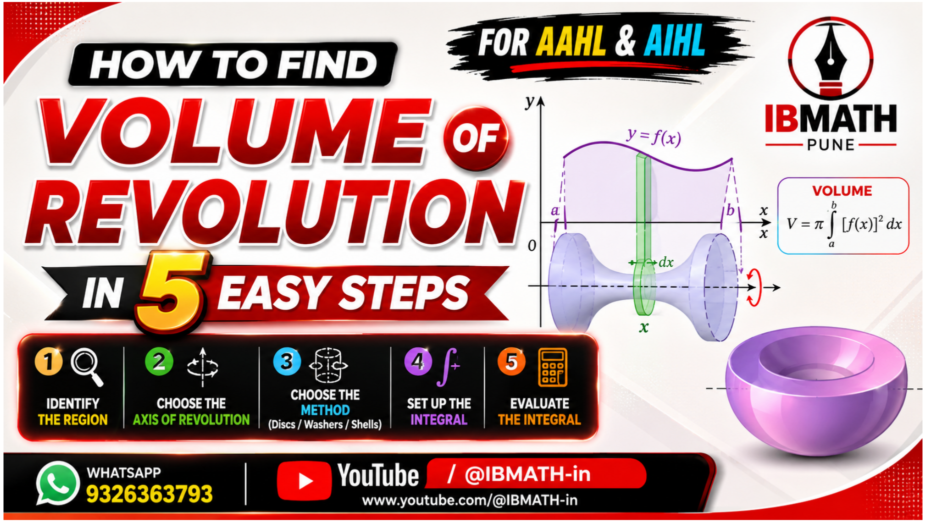
$$A = \int_0^{\pi} c_1 - c_2 dx$$

$$= \int_0^{\pi} (4\sin x - \sin x)\, dx$$

$$= \int_0^{\pi} 3\sin x dx$$

$\therefore A = 6$ unit² (using G. D. C.)

Practice questions PDF link is given in the description of this video.



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